

NFL Defensive Coverage Predictor

CS/SE 4AL3 – Project Proposal

Group 77

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Task Overview and Significance

Our task is to recognize defensive coverages before the snap. This is challenging because defenses often try to disguise their coverage by showing one formation or alignment before the snap and executing another after the snap. This makes it difficult for an offensive player or coordinator to determine the defense's true coverage based only on its pre-snap alignment.

This matters because accurately identifying defensive coverage before the play begins could help offensive coordinators and quarterbacks make better pre-snap decisions, such as adjusting the play, route combinations, or blocking scheme.

This project is non-trivial because the model must make its prediction using only pre-snap information, while accounting for defensive disguises, player positioning, and different formations. The novel aspect of our project is developing a machine learning approach specifically focused on predicting the coverage before the snap, rather than using information from the play after it has already developed. We will also investigate which pre-snap features are most useful for distinguishing between different defensive coverages.

Task Definition

Input/data type: [e.g., text, images, tabular records]

Prediction task: [classification / regression / generation]

Labels and output: [number of classes; label definitions; single-label or multi-label; or regression target]

Unit of prediction: [What one data point represents]

[State the task precisely, including the model inputs and desired outputs.]

Data Sources and Collection Plan

[Identify each source and provide direct links. Explain how data will be obtained (API, permitted scraping, open dataset, etc.), terms of service, access, and rate limits. Describe collection, annotation/labeling effort, preprocessing, and available metadata. Say whether you will use all data or a subset/features and why. For Kaggle, include both the Kaggle page and the original source. Do not use datasets with no source information.]

Expected dataset size: [at least 1,000 data points; justify any exception]

Metadata: [Available fields and their relevance]

Three example data points with labels

1. **Input:** [Example 1] **Label:** [Label]

2. **Input:** [Example 2] **Label:** [Label]

3. **Input:** [Example 3] **Label:** [Label]

Proposed Solution and Evaluation

[Describe features and target labels, candidate models, and how you will approach the problem. Do not propose simple linear regression. Discuss existing solutions and how your work differs. Explain evaluation design, metrics, data splits/baselines, and how you will decide whether the model is good. Name intended libraries.]

Planned libraries: [e.g., scikit-learn, PyTorch, pandas]

Evaluation: [Metrics, baseline(s), validation plan, and success criteria]

Sources inspiring the approach (2–5)

, URL or DOI

, URL or DOI

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TA Consultation

We consulted with Abdelaziz Serour twice throughout the development of our project idea. During our first interaction, our team proposed developing a SpecDec system for tool calls. However, after discussing the idea with the TA, we concluded that it was too complex and theoretical for the scope of the project. As a result, we decided to pivot toward a more practical and applied idea.

We then developed the idea of recognizing defensive coverages during the pre-snap period. We reached out to Abdelaziz again through Outlook to discuss whether this idea was feasible, and he agreed that it was a suitable project that we could proceed with.

Team Contract

Write this section yourselves. Do not use generative AI to write the proposal or team contract.

Team mission: [What is the purpose of the team?]

Roles, duties, and expectations: [Describe each member’s responsibilities and shared expectations. Focus on how you will work together.]

Leadership and coordination: [How will you organize meetings, make decisions, track work, and share facilitation?]

Communication and conflict resolution: [How will you raise concerns, meet, discuss disagreements, and resolve conflict? Add any other useful ground rules.]

Signatures

Printed name	Signature	Date
Justin Kwinecki	_____	October 7, 2026
Zayed Umer	_____	October 7, 2026
William Pol	_____	October 7, 2026